Customer Stories / Life Sciences / USA

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Amgen Scientists Use AWS HealthOmics to Reduce the Time and Cost of Therapeutics Analysis

Learn how Amgen, a leading biopharmaceutical company, standardized and simplified access to cloud-based epigenomics analysis pipelines for scientists using AWS HealthOmics.

Overview | Opportunity | Solution | Outcome | AWS Services Used

Off-loaded infrastructure management

for omics pipelines in the US with a fully managed infrastructure

Up to 40%

overall cost savings

Empowered

self-service among scientists with a sir

Shortened

time to develop and run omics pipelines, with transparent debugging for rapid troubleshooting

Improved

> pipeline robustness and reproducibility

Overview

Omics data analysis—which involves sequencing biological molecules such as DNA, RNA, and/or proteins—holds great promise for improving health outcomes and accelerating scientific discovery. But analyzing large, complex, and high-dimensional datasets presents challenges and requires specialized tools to support high-throughput computing and petabytes of storage.

Amgen, a leader in biotechnology, recognized the importance of omics data to therapeutic and scientific breakthroughs. Amgen researchers in the US initially used Genomics Workflows (GWF) on Amazon Web Services (AWS) to support Amgen omics workloads. But accessing the GWF through a command-line interface required high technical acumen, and scientists in Research and Development had to rely on a busy IT team to manage and debug the required compute resources, causing delays to workflow deployments in the GWF.

To simplify and standardize the process of deriving insights from omics data, Amgen adopted AWS HealthOmics, which helps healthcare and life sciences organizations to store, query, and analyze genomic, transcriptomic, and other omics data at scale. Using AWS HealthOmics, Amgen centralized its omics analysis pipelines, provided a simplified interface for launching and tracking jobs, which has the potential to cut overall costs by 25–40 percent compared with the prior solution—all while speeding up cycle times. Using the secure storage workflows of AWS HealthOmics, Amgen accelerated its omics pipelines for faster cycle times t insights and discoveries.

Opportunity | Using AWS HealthOmics to Standardize Data Pipelines for Amgen

Founded in 1980 in California, Amgen seeks to unlock the potential of biology and help patients suffering from serious illnesses by discovering, developing, manufacturing, and delivering innovative therapeutics. Its R&D team leverages a deep understanding of target and disease biology to find new and better ways to defeat the world’s toughest diseases, and uses omics analysis for multiple stages in the drug development process–from target discovery to protein manufacturing.

Realizing the benefits of the cloud for running genomics workflows, Amgen in 2019 implemented GWF on AWS and began to migrate its omics pipelines from its on-premises implementation to AWS. But the AWS solution—although powerful—was not fully self-service. An IT support team had to set up pipelines for scientists, and it was difficult to troubleshoot and track tasks or debug runs. In addition, scientists were leveraging multiple compute environments to run analyses, taxing the infrastructure team and making it difficult to view and track technological resources.

In October 2022, AWS shared with Amgen a pilot opportunity to test AWS HealthOmics. Amgen recognized this was an opportunity to standardize and centralize its omics workflows while giving scientists robust and customizable pipeline processing options. “We really appreciated the work that our teams had done to create pipelines and get them up and running on the GWF,” says Ittai Eres, Senior Scientist in Bioinformatics Technologies at the Center for Research Acceleration by Digital Innovation within Amgen. “But we knew that using AWS HealthOmics would be a faster process and help us standardize and share pipelines across different analysts.

Solution | Accelerating Turnaround Time for Omics Workloads While Facilitating Access for Scientists

Amgen began to implement AWS HealthOmics for five different pipelines, relieving the IT team of time-consuming infrastructure setup and management. “It freed up our time from constantly tweaking compute environments, pipelines and queues to make sure that everything was working properly, and it provided us the scale and stability we previously did not have,” says Jamie Rosner, Technical Product Manager for the Omics Team in Research and Development Informatics. “Instead, we can now spend that time developing the pipelines themselves.”

Using AWS HealthOmics, Amgen stores and manages workflows in one place with the same a el, providing scalability that the previous solution lacked. Amgen also maintains its chain of data the pipeline and has access to a full audit trail, helping with compliance requirements. “Using AWS HealthOmics, we can further our understanding of biology, target, and disease, ultimately informing decisions in Research and Development," says Eres. "For a company of Amgen's size, having everything in one place is tremendous for reproducibility, transparency, security, and permissions."

Using AWS HealthOmics, scientists have immediate visibility into all available pipelines. Additionally, AWS HealthOmics supports Amgen's requirements with Nextflow workflows while offering the flexibility to expand to other languages like WDL or CWL if needed, without having to reinvent the wheel. Scientists can access AWS HealthOmics through a simple graphical user interface, without needing command-line expertise. "This is something that we lacked before," says Rosner. "Now, scientists with differing levels of technical skill have the same ability to analyze various types of data in a simple point-and-click format." Scientists also can better understand the effectiveness of different types of analyses, such as gaining greater visibility into the benefits of the "Cleavage Under Targets and Release Using Nuclease" (CUT&RUN) method, a relatively new type of omics analysis that isolates specific protein-DNA complexes, enabling profiling of the epigenome. "Now it's feasible to rapidly port it into AWS HealthOmics and see if this is a new data type that we are going to be interested in using," says Eres. "In the past, it would have been a headache even to assess the utility of a new omics profiling strategy, which may have resulted in deprioritization and missing out on valuable biological insights."

By centralizing its resources, Amgen reduced unnecessary costs related to redundant software or duplicate licensing. Through close collaboration with AWS, Amgen provided valuable product feedback to meet its needs while ultimately driving enhancements that benefited other AWS HealthOmics customers.

AWS HealthOmics can also easily integrate with other workflows and systems. For example, scientists can send the metadata that is available through their laboratory information management system (LIMS) directly to AWS HealthOmics through an API call.

Outcome | Bringing Medicines to Market Faster through Streamlined Decision-Making

Using tools such as AWS HealthOmics, Amgen is bringing medicines to patients faster, by better understanding targets and assays across all stages of the drug development pipeline. For example, analysis of CUT&RUN data across many different samples helped wet lab scientists decide which antibodies to pursue and which were not specific enough to the desired target. The company is going live with its first production workload in mid-2024, with more than 30 additional workloads scheduled to migrate to the cloud for cost and shareability.

Amgen is also connecting the workflows to various internal systems such as LIMS to facilitate cross-system metadata management, automated run parameter generation, and automated run start. "Our vision is to build a truly complete automated system. As soon as the data are sequenced, they automatically run and are analyzed so that scientists arriving in the morning are ready to begin their work with minimal human intervention," says Eres.

Amgen scientists in the US will continue to further democratize the use of omics, sharing work done across the organization in a user-friendly way and building a central resource for omics data analysis. "For scientists interested in broadening accessibility, usage, and visibility of omics pipelines, AWS HealthOmics is a uniquely appealing and fit-for-purpose solution," says Eres. "We answer questions more quickly and assess much more rapidly whether a target deserves attention. Ultimately, we hope to get medicines more quickly to the people that need them."

About Amgen

Amgen discovers, develops, manufactures, and delivers innovative medicines to help millions of patients in their fight against some of the world's toughest diseases. For more information, visit Amgen.com.

AWS Services Used

AWS HealthOmics

AWS HealthOmics is a purpose-built service that helps healthcare and life science organizations and their software partners store, query, and analyze genomic, transcriptomic, and other omics data and then generate insights from that data to improve health

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